CIMAT

Homework 5

An Application of Takahama's Constrained Optimization Technique

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1. Introduction

Constrained optimization deals with problems in which we seek the solution to the problem

$$\min_{x \in \mathbb{R}^d} f(x)$$

subject to the following sets

$$G = \left\{x \in \mathbb{R}^d : g_i(x) = c_i\right\}_{i=1}^n$$
$$H = \left\{x \in \mathbb{R}^d : h_j(x) \leq d_j\right\}_{j=1}^m$$

One can solve these problems using analytics methods, such as Lagrange multipliers or KKT conditions. However, one may also take a stochastic approach to solve the problem. One may observe that the set G has measure 0 in $\mathbb{R}^d$, thus unreasonable to expect a stochastic process to land exactly where our equality constraint asks us to.

To bypass this problem, we replace the equality constraints by inequalities, essentially allowing for an ε error around each c_i. That is,

$$G_\varepsilon = \left\{x \in \mathbb{R}^d : \|g_i(x)\| \leq \varepsilon\right\}_{i=1}^n$$

This new set has positive measure, so we can expect a random process to satisfy this condition with positive probability.

In this document, we perform the analytic calculations of the method.

2. Takahama Method in $\mathbb{R}^2$

Let $x \in \mathbb{R}^d$ be a feasible point. The iteration depends on the violations.

1. If only 1 restriction $h_{i(x)} > 0$ is violated: we move in the opposite direction of the gradient of h_i. That is,

$$\Delta x = -\frac{h_i(x)}{\|\nabla h_i(x)\|^2} \nabla h_i(x)$$

2. If both restrictions are violated: the step is given by

$$\Delta x = -J^{-1} V,$$

where

$$J = \begin{bmatrix} \nabla h_1(x)^\top \\ \nabla h_2(x)^\top \end{bmatrix}$$

and

$$V = \begin{bmatrix} h_1(x) \\ h_2(x) \end{bmatrix}.$$

3. Example

Let the objective function be

$$f(x, y) = \left(x^2 + y - 11\right)^2 + \left(x + y^2 - 7\right)^2$$

subject to

$$\begin{aligned} h_1(x, y) &= 26 - (x - 5)^2 - y^2 \geq 0, \\ h_2(x, y) &= 20 - 4x - y \qquad\quad \geq 0. \end{aligned}$$

We reverse the inequalities to obtain the following restrictions

$$\begin{aligned} h_1(x, y) &= y^2 + (x - 5)^2 - 26 \leq 0, \\ h_2(x, y) &= y + 4x - 20 \qquad\quad \leq 0. \end{aligned}$$

Now, the violation occurs when $g_{i(x,y)} > 0$. The gradients of the restrictions are given by

$$\begin{aligned} \nabla h_1(x, y) &= \begin{pmatrix} 2(x - 5) \\ 2y \end{pmatrix} \\ \nabla h_2(x, y) &= \begin{pmatrix} 4 \\ 1 \end{pmatrix} \end{aligned}$$

3.1. Point 1

We use start at the point $P = (10, 5)$. We evaluate the restrictions at this point, and obtain the following:

$$\begin{aligned} h_1(10, 5) &= (10 - 5)^2 + 5^2 - 26 = 24 > 0 \\ h_2(10, 5) &= 4(10) + 5 - 20 \qquad\quad = 25 > 0 \end{aligned}$$

Thus, both restriction are violated. The gradients are given by

$$\nabla h_1(10,5) = \begin{pmatrix} 10 \\ 10 \end{pmatrix}$$

$$\nabla h_2(10,5) = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

Therefore, the Jacobian evaluates to

$$|J| = \begin{vmatrix} 10 & 10 \\ 4 & 1 \end{vmatrix} = -30,$$

and thus its inverse is

$$J^{-1} = -\frac{1}{30}\begin{pmatrix} 1 & -10 \\ -4 & 10 \end{pmatrix}$$

so

$$\Delta x = -\begin{pmatrix} -\frac{1}{30} & \frac{1}{3} \\ \frac{4}{30} & -\frac{1}{3} \end{pmatrix}\begin{pmatrix} 24 \\ 25 \end{pmatrix} = \frac{1}{15}\begin{pmatrix} -113 \\ 77 \end{pmatrix} \approx \begin{pmatrix} -7.533 \\ 5.133 \end{pmatrix}$$

Our next evaluation point is then

$$P_k = P_{k-1} + \Delta x = \begin{pmatrix} 10 \\ 5 \end{pmatrix} + \begin{pmatrix} -7.533 \\ 5.133 \end{pmatrix} = \begin{pmatrix} 2.467 \\ 10.133 \end{pmatrix}$$

3.2. Point 2

Let $P = (5,5)$, then

$$\begin{aligned} h_1(5,5) &= (5-5)^2 + 5^2 - 26 = -1 \leq 0 \\ h_2(5,5) &= 4(5) + 5 - 20 \qquad\quad = 5 > 0 \end{aligned}$$

Thus, only h_2 is violated, so we move in the opposite direction of its gradient.

$$\nabla h_2(5,5) = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \Longrightarrow \|\nabla h_2\|^2 = 17.$$

Therefore,

$$\Delta x = -\frac{5}{17}\begin{pmatrix} 4 \\ 1 \end{pmatrix} \approx \begin{pmatrix} -1.176 \\ -0.294 \end{pmatrix}$$

and the new point is

$$P_k = P_{k-1} + \Delta x = \begin{pmatrix} 5 \\ 5 \end{pmatrix} + \begin{pmatrix} -1.176 \\ -0.294 \end{pmatrix} = \begin{pmatrix} 3.824 \\ 4.706 \end{pmatrix}$$

3.3. Point 3

Let $P = (2, 8)$, then

$$h_1(2, 8) = 47 > 0$$
$$h_2(2, 8) = -4 \leq 0$$

Thus, only h_1 is violated, so we move in the opposite direction of its gradient.

$$\nabla h_1(2, 8) = \begin{pmatrix} -6 \\ 16 \end{pmatrix} \Longrightarrow \|\nabla h_1\|^2 = 292.$$

Therefore,

$$\Delta x = -\frac{47}{292}\begin{pmatrix} -6 \\ 16 \end{pmatrix} \approx \begin{pmatrix} 0.966 \\ -2.575 \end{pmatrix}$$

and the new point is

$$P_k = P_{k-1} + \Delta x = \begin{pmatrix} 2 \\ 8 \end{pmatrix} + \begin{pmatrix} 0.966 \\ -2.575 \end{pmatrix} = \begin{pmatrix} 2.966 \\ 5.425 \end{pmatrix}$$

3.4. Point 4

Let $P = (8, 8)$, then

$$h_1(8, 8) = 47 > 0$$
$$h_2(8, 8) = 20 > 0$$

Both restrictions are violated. Thus, the Jacobian is

$$|J| = \begin{vmatrix} 6 & 16 \\ 4 & 1 \end{vmatrix} = -58$$

and thus

$$J^{-1} = -\frac{1}{58}\begin{pmatrix} 1 & -16 \\ -4 & 6 \end{pmatrix}.$$

Therefore,

$$\nabla x = \frac{1}{58}\begin{pmatrix} -273 \\ -68 \end{pmatrix} \approx \begin{pmatrix} -4.707 \\ -1.172 \end{pmatrix}$$

and the new point is

$$P_k = P_{k-1} + \Delta x = \begin{pmatrix} 8 \\ 8 \end{pmatrix} + \begin{pmatrix} -4.707 \\ -1.172 \end{pmatrix} = \begin{pmatrix} 3.293 \\ 6.828 \end{pmatrix}.$$

3.5. Graph

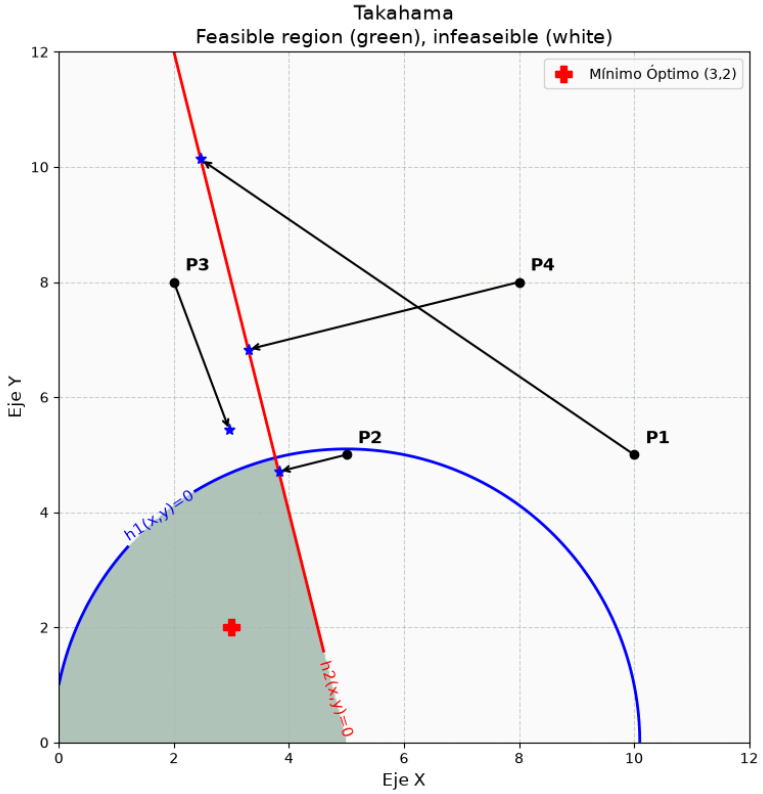


Figure 1: Plot of the restrictions and where the points end up after one iteration of the Takahama process. We observe that they generally approach the feasible lines.